UNIT V – NUCLEAR ENERGY SOURCE

Syllabus: General Properties of Nucleus (Size, Mass, Density, Charge) – Mass Defect – Binding Energy - Disintegration in fission –Nuclear Reactor: Materials Used in Nuclear Reactors. – PWR – BWR – FBTR. Nuclear fusion reactions for fusion reactors-D-D and D-T reactions, p-p and C-N cycle) Basic principles of Nuclear Fusion reactors.

UNIT – V NUCLEAR ENERGY SOURCE

5.1. INTRODUCTION

The atomic nucleus was discovered by Earnest Rutherford in 1911. **The nucleus** consists of an elementary particles, protons and neutrons which are known as nucleons. A proton has positive charge of the same magnitude as that of electron and its rest mass is about 1836 times the mass of an electron. A neutron is electrically neutral, whose mass is almost equal to the mass of the proton. The nucleons inside the nucleus are held together by strong attractive forces called nuclear forces. A nucleus of an element is represented as $_ZX^A$, where X is the chemical symbol of the element. Z represents the atomic number which is equal to the number of protons and A, the mass number which is equal to the total number of protons and neutrons. The number of neutrons is represented as N which is equal to A−Z. For example, the chlorine nucleus is represented as $_{17}Cl^{35}$. It contains 17 protons and 18 neutrons.

5.2. CLASSIFICATION OF NUCLEI

(i). Isotopes

Isotopes are atoms of the same element having the same atomic number Z but different mass number A. The nuclei $_1H^1$, $_1H^2$ and $_1H^3$ are the isotopes of hydrogen. In other words isotopes of an element contain the same number of protons but different number of neutrons.

(ii). Isobars

Isobars are atoms of different elements having the same mass number A, but different atomic number Z. The nuclei $_8O^{16}$ and $_7N^{16}$ represent
isobars. Since isobars are atoms of different elements, they have different physical and chemical properties.

(iii). Isotones

Isotones are atoms of different elements having the same number of neutrons $_6C^{14}$ and $_8O^{16}$ are some examples of isotones.

5.3. General properties of nucleus

(i). Nuclear size

If the nucleus is assumed to be spherical, an empirical relation is found to hold good between the radius of the nucleus R and its mass number A. It is given by

$$R \propto A^{1/3}$$

$$R = r_o A^{1/3} \qquad \ldots(1)$$

where r_o is the constant of proportionality and is equal to 1.3F (1 Fermi, F = 10^{-15} m)

(ii). Nuclear density

The nuclear density ρ_N can be calculated from the mass and size of the nucleus.

$$\rho_N = \frac{\text{Nuclear mas}}{\text{Nuclear volume}}$$

Nuclear mass = Am_N where, A = mass number and m_N = mass of one nucleon and is approximately equal to 1.67×10^{-27} kg.

$$\text{Nuclear volume} = \frac{4}{3}\pi R^3$$

$$= \frac{4}{3}\pi\,(r_o\,A^{1/3})$$

$$= \frac{Am_N}{\frac{4}{3}\pi\,(r_0A^{1/3})^3} \qquad \ldots(1)$$

$$\therefore \quad \rho_N = \frac{m_N}{\frac{4}{3}\pi\,r_0^{\,3}}$$

Substituting the known values, the nuclear density is calculated as 1.816×10^{17} kg m^{-3} which is almost a constant for all the nuclei irrespective of its size.

(iii). Nuclear charge

The charge of a nucleus is due to the protons present in it. Each proton has a positive charge equal to 1.6×10^{-19} C.

∴ The nuclear charge = Ze, where Z is the atomic number.

(iv). Nuclear mass

As the nucleus contains protons and neutrons, the mass of the nucleus is assumed to be the mass of its constituents. Assumed nuclear mass = $Zm_P + Nm_n$, where m_p and m_n are the mass of a proton and a neutron respectively. However, from the measurement of mass by mass spectrometers, it is found that the mass of a stable nucleus (m) is less than the total mass of the nucleons. i.e., mass of a nucleus, $m<(Zm_p+Nm_n)$, $Zm_p + Nm_n - m = \Delta m$, where Δm is the mass defect .

(v) Spin angular momentum

Both the proton and neutron, like the electron, have an intrinsic spin. The spin angular momentum is computed by $L_s = \frac{l(l+1)h}{2\pi.}$

(vi). Resultant angular momentum

In addition to the spin angular momentum, the protons and neutrons in the nucleus have an orbital angular momentum. The resultant angular momentum of the nucleus is obtained by adding the spin and orbital angular momentum of all the nucleons within the nucleus. This total angular momentum is called nuclear spin.

5.4. Atomic mass unit

It is convenient to express the mass of a nucleus in atomic mass unit (amu), though the unit of mass is kg. (One atomic mass unit is considered as one twelfth of the mass of carbon atom ${}_6C^{12}$). Carbon of atomic number 6 and mass number 12 has mass equal to 12 amu.1 amu = 1.66×10^{-27} kg

The energy equivalence of 1 amu can be calculated in electron-volt:

Einstein's mass energy relation is, $E = mc^2$

Here, $m = 1 \text{ amu} = 1.66 \times 10^{-27}$ kg

$c = 3 \times 10^8 \text{ ms}^{-1}$

$$E = mc^2$$

$$\therefore E = 1.66 \times 10^{-27} \times (3 \times 10^8)^2 \text{ J}$$

One electron-volt (eV) is defined as the energy of an electron when it is accelerated through a potential difference of 1 volt.

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ coulomb} \times 1 \text{ volt}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ joule}$$

$$1 \text{ joule} = \frac{1}{1.6 \times 10^{-19}} eV$$

$$E = \frac{1.66 \times 10^{-27} \times (3 \times 10^8)^2}{1.6 \times 10^{-19}} eV$$

$$E = 931 \times 10^6 \text{eV}$$

$$= 931 \text{ Million electron Volt} = 931 \text{ MeV}$$

Thus, energy equivalent of 1 amu = 931 MeV

5.5. Mass defect

The difference in the total mass of the nucleons and the actual mass of the nucleus is known as the mass defect.

$$Zm_p + Nm_n - m = \Delta m$$

5.6. Binding energy

When the protons and neutrons combine to form a nucleus, the mass that disappears (mass defect, Δm) is converted into an equivalent amount of energy (Δmc^2). This energy is called the binding energy of the nucleus.

$$\therefore \text{Binding energy} = [Zm_P + Nm_n - m]\, c^2 = \Delta m\, c^2$$

The binding energy of a nucleus determines its stability against disintegration. In other words, if the binding energy is large, the nucleus is stable and vice versa.

5.6.1. Binding energy per nucleon

The binding energy per nucleon is $\left(\frac{B}{A}\right) = \frac{\text{Binding energy of the nucleus}}{\text{Total number of nucleons}}$

5.7. Packing fraction

The ratio between the mass defect *(Δm)* and the mass number (A) is called the packing fraction *(f)*

$$f = \frac{\Delta m}{A} \qquad \ldots(1)$$

Packing fraction means the mass defect per nucleon. Packing fraction is a measure of the comparative stability of the atom.

$$\text{Packing fraction} = \frac{\text{Isotopic mass} - \text{Mass number}}{\text{Mass number}}$$

Packing fraction may have a positive or negative sign. If packing fraction is negative, the isotopic mass is less than the mass number. In such cases, some mass gets transformed into energy in the formation of that nucleus. A positive packing fraction would imply a tendency towards instability.

5.8. Variation of Binding energy per nucleon with mass number

It is found that the binding energy per nucleon varies from element to element. A graph is plotted with the mass number of the nucleus along the X−axis and binding energy per nucleon along the Y-axis as shown in the figure 5.1.

(i) The binding energy per nucleon increases sharply with mass number A upto 20. It increases slowly after A = 20. For A<20, there exists recurrence of peaks corresponding to those nuclei, whose mass numbers are multiples of four and they contain not only equal but also even number of protons and neutrons. Example: ${}_2He^4$, ${}_4Be^8$, ${}_6C^{12}$, ${}_8O^{16}$, and ${}_{10}Ne^{20}$. The curve becomes almost flat for mass number between 40 and 120. Beyond 120, it decreases slowly as A increases.

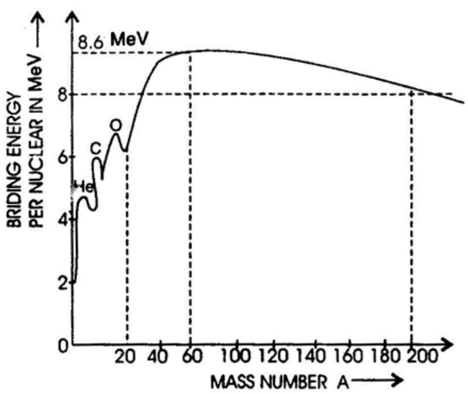

Fig.5.1. Variation of Binding per nucleon as a function of A

(ii) The binding energy per nucleon reaches a maximum of 8.6MeV at A=56, corresponding to the iron nucleus ($_{26}Fe^{56}$). Hence, iron nucleus is the most stable.

(iii) The average binding energy per nucleon is about 8.5MeV for nuclei having mass number ranging between 40 and 120. These elements are comparatively more stable and non radioactive.

(iv) For higher mass numbers the curve drops slowly and the $\frac{BE}{A}$ is about 7.6 MeV for Uranium. Hence, they are unstable and radioactive.

(v) The lesser amount of binding energy for lighter and heavier nuclei explains nuclear fusion and fission respectively. A large amount of energy will be liberated if lighter nuclei are fused to form heavier one (fusion) or if heavier nuclei are split into lighter ones (fission).

5.9. Nuclear fission

The process of breaking up of the nucleus of a heavier atom into two fragments with the release of large amount of energy is called **nuclear fission**. The fission is accompanied by the release of neutrons. The fission reactions with $_{92}U^{235}$ are represented as

$$_{92}U^{235} + {_0n^1} \rightarrow {_{92}U^{236}} \rightarrow {_{56}Ba^{141}} + {_{36}Kr^{92}} + 3\,{_0n^1} + Q(\text{energy}) \qquad \ldots(1)$$

$$_{92}U^{235} + {_0n^1} \rightarrow {_{54}Xe^{140}} + {_{38}Sr^{94}} + 2\,{_0n^1} + Q(\text{energy}) \qquad \ldots(2)$$

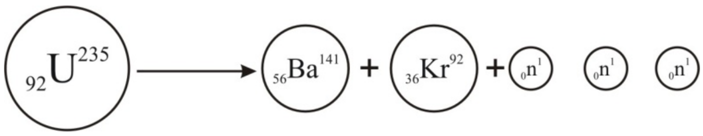

Fig. 5.2 Fission in Uranium

In the above examples the fission reaction is taking place with the release of 3 neutrons and 2 neutrons.

5.9.1 Energy released in fission

Let us calculate the amount of energy released during the fission of $_{92}U^{235}$ with a neutron. The fission reaction is

$$_{92}U^{235} + {_0n^1} \rightarrow {_{92}U^{236}} \rightarrow {_{56}Ba^{141}} + {_{36}Kr^{92}} + 3\,{_0n^1} + Q$$

Mass of $_{92}U^{235}$ = 235.045733 amu, Mass of $_0n^1$ = 1.008665 amu

Total mass of the reactant = 236.054398 amu

Mass of $_{56}Ba^{141}$ = 140.9177 amu

Mass of $_{36}Kr^{92}$ = 91.8854 amu

Mass of 3 neutrons = $(3 \times 1.008665) = 3.025995$ amu

Total mass of the products = 235.829095 amu

$\therefore$ Mass defect = 236.054398 – 235.829095 = 0.225303 amu

As, 1 amu = 931 MeV, energy released in a fission

$= 0.225303 \times 931\text{MeV} = 200$ MeV

5.10. Chain reaction

A chain reaction is a self propagating process in which the number of neutrons goes on multiplying rapidly almost in a geometrical progression during fission till whole of fissile material is disintegrated.

5.10.1. Effective multiplication factor

It is quantity to consider the reactor control. This factor determines whether the chain reaction will continue at a steady rate, increase or decrease. It is defined as the ratio of production of neutrons 'P' to the combined rate of absorption 'A' and rate of leakage of 'L' of neutrons.

$$K_e = \frac{\text{production of Neutrons (P)}}{\text{Combined Rate of absorption (A)} + \text{Rate of leakage of Neutrons (L)}} = \frac{P}{A+L}$$

If $K_e = 1$, chain reaction is critical

$K_e > 1$, chain reaction is building up i.e., super critical

$K_e < 1$, chain reaction is dying down i.e., sub critical

The chain reaction may be classified into two types

1. Controlled chain reaction
2. Uncontrolled chain reaction

5.10.2. Controlled chain reaction

In the controlled chain reaction the number of fission producing neutron is kept constant and is always equal to one. The reaction is sustained in a controlled manner. Controlled chain reaction is taking place in a nuclear reactor. When a thermal neutron bombards U^{235} nucleus, it breaks into two fission fragments and three fast neutrons as shown in Fig. 5.3.

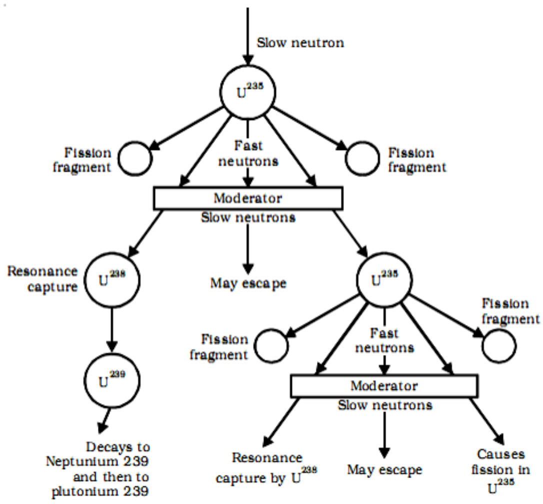

Fig.5.3. Controlled chain reaction

One neutron may escape and one neutron may be captured by U^{238} which decays to Np^{239} and then to Pu^{239}. One neutron is available for carrying out chain reaction. The chain reaction is possible, only when the loss of neutrons is less than the neutrons produced.

5.10.3. Uncontrolled Chain reaction

In the uncontrolled chain reaction, the number of neutrons multiplies indefinitely and the entire amount of energy is released within a fraction of a second. Such reactions are taking place in an atom bomb as shown in Fig. 5.4. A chain reaction can be maintained, if an each nucleus undergoing fission, at least one neutron is produced which causes the fission of another nucleus. This condition is conveniently expressed in terms of a multiplication factor *K*.

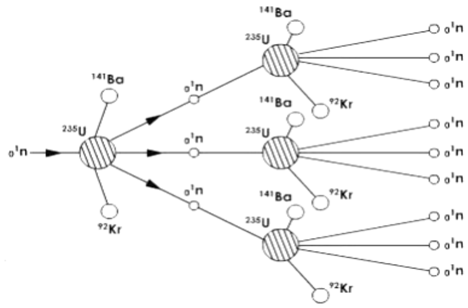

Fig.5.4. Uncontrolled chain reaction

5.10.4. Multiplication factor (*k*)

The ratio of secondary neutrons produced to the original neutrons is called the multiplication factor.

$$K = \frac{\text{Number of neutrons in any one generation}}{\text{Number of neutrons in the preceeding generation}}$$

The chain reaction will be critical or steady when k=1

Building up or supercritical when k>1

Dying down or subcritical when k<1

5.10.5. Critical size and Critical mass

Critical size of a system containing a fissile material is defined as the minimum size in which at least one neutron is available for further fission reaction. The mass of the fissile material at the critical size is called **critical mass**. The chain reaction is not possible if the size is less than the critical size.

5.10.6. Fissile and Fissionable Nuclides and Fertile materials

The species like $_{92}U^{235}$, $_{92}U^{233}$ and $_{94}Pu^{239}$ which undergo fission by neutron's energy of almost above zero MeV, are called **fissile nuclides** whereas nuclides like $_{90}Th^{233}$, $_{92}U^{238}$, which include fission by neutron's energy of greater than 1 MeV, are called **fissionable nuclides**. Since $_{90}Th^{232}$ and $_{92}U^{238}$ both can be converted into fissile material (like $_{92}U^{239}$, $_{92}U^{233}$ and $_{94}Pu^{239}$), hence they are also called as **fertile materials**.

5.10.7. Fast Neutrons and Thermal Neutrons

The neutrons that leave the fission site typically have kinetic energy levels of about 2 MeV. These are referred to as **Fast Neutrons**. Eventually, these neutrons give up this kinetic energy (primarily through elastic scattering) and come to thermal equilibrium with their surroundings. Such slow neutrons are called **Thermal Neutrons**. A thermal neutron has an energy level of approximately 0.025 eV.

5.10.8. Prompt and Delayed Neutrons

There are two sources of neutrons in a reactor. Most of the neutrons from fission are ejected at the instant of fission. These are called **prompt neutrons**. In addition, a small number of the fission fragments decay by neutron emission. The average time from fission to neutron emission by the fragment is thirteen seconds. These are called **delayed neutrons**.

5.11. Nuclear fusion

Nuclear fusion is a process in which two or more lighter nuclei combine to form a heavier nucleus with the release of energy. So the nuclear fusion reaction is an exothermic nuclear reaction with positive Q-value. This is also called thermo nuclear reaction. During fusion reaction the mass of the product nucleus is always less than the sum of the masses of the individual lighter nuclei. The difference in mass is converted into energy. The fusion process can be carried out only at an extremely high temperature of the order of 10^7 K because only at these very high temperatures the nuclei are able to overcome their mutual repulsion. Therefore before fusion, the lighter nuclei must have their temperature raised by several million degrees. So the nuclear fusion reactions are known as thermo-nuclear reactions.

5.11.1. Nuclear fusion in the Sun and the Stars

Fusion is the source of stellar energy. It has been estimated that the total energy radiated by sun is about 3.8×10^{26} joule per second. The origin of such a tremendous amount of energy is the fusion of protons into helium. Proton–proton cycle and carbon–nitrogen cycle are the two important types in which nuclear fusion takes place in sun and stars.

5.11.2. Proton-Proton cycle

The production of stellar energy is by thermo nuclear reactions in which protons are continuously transformed into helium nuclei. The proton-proton cycle takes place in **sun.**

$${}_1H^1 + {}_1H^1 \rightarrow {}_1H^2 + {}_{+1}e^0 + \nu \text{ (emission of positron and neutrino)} \quad \ldots\ldots(1)$$

$${}_1H^1 + {}_1H^2 \rightarrow {}_2He^3 + \gamma \text{ (emission of gamma rays)} \quad \ldots\ldots (2)$$

$$2\,{}_2He^3 \rightarrow {}_2He^4 + 2\,{}_1H^1 \quad \ldots\ldots (3)$$

The reaction cycle is written as

$$4\,{}_1H^1 \rightarrow {}_2He^4 + 2\,{}_1e^0 + 2\nu + \text{energy (26.7 MeV)} \quad \ldots(4)$$

Thus four protons fuse together to form an alpha particle and two positrons with a release of large amount of energy.

5.11.3. Carbon-Nitrogen cycle

The carbon-nitrogen cycle takes place in **stars**. The following cycle of reactions take place in carbon – nitrogen cycle in which carbon acts as a catalyst.

$_1H^1 + {}_6C^{12} \rightarrow {}_7N^{13*} + \gamma$ (emission of gamma rays) …..(1)

$_7N^{13*} \rightarrow {}_6C^{13} + {}_1e^0 + \nu$ (emission of positron and neutrino)…..(2)

$_1H^1 + {}_6C^{13} \rightarrow {}_7N^{14} + \gamma$ (emission of gamma rays) …..(3)

$_1H^1 + {}_7N^{14} \rightarrow {}_80^{15*} + \gamma$ (emission of gamma rays) …..(4)

$_80^{15*} \rightarrow {}_7N^{15} + {}_1e^0 + \nu$ (emission of positron and neutrino) …..(5)

$_1H^1 + {}_7N^{15} \rightarrow {}_6C^{12} + {}_2He^4$ …..(6)

The overall reaction of the above cycle is given as

$4\ {}_1H^1 \rightarrow {}_2He^4 + 2{}_1e^0 + 2\nu$ + energy (26.7 MeV) …(7)

5.12. Nuclear Fusion on Earth

Man-made nuclear fusion uses a different reaction to that which occurs in the stars. Fusion on Earth starts by fusing Deuterium and Tritium. (Tritium is an isotope of Hydrogen which contains 1 proton and 2 neutrons).

5.12.1. Making tritium

The fusion of Deuterium and Tritium reaction yields 17.6 MeV of energy but requires a temperature of approximately 40 Million Kelvin to overcome the coulomb barrier and ignite it. The Deuterium fuel is abundant; the problem with the D-T process is that Tritium is not found naturally on earth. It has to be manufactured in a nuclear laboratory. One method is to use the Lithium isotope, $_3Li^6$, and the following reaction:

$_1n^0 + {}_3Li^6 = {}_2He^4 + {}_1H^3 + 4.8MeV$ …(1)

$_1n^0(fast) + {}_3Li^7 = {}_2He^3 + {}_1H^3 + {}_1n^0(slow)$ …(2)

5.12.2. Deuterium – Tritium Cycle of Fusion

When two deuterons come close together they may just bounce off each other. The four fusion reactions which can occur with Deuterium and Tritium can be considered to form a Deuterium-Tritium cycle. The four reactions:

(Deuterium – deuterium fusion)

$_1H^2 + {}_1H^2 = {}_2He^3 + {}_on^1 + 3.27MeV$ …..(1)

$_1H^2 + {}_1H^2 = {}_1H^3 + {}_1H^1 + 4.03MeV$ ….(2)

(Deuterium – Tritium fusion)

$_1H^2 + {}_1H^3 = {}_2He^4 + {}_on^1 + 17.59MeV$ ….(3)

(Deuterium – Helium-3 fusion)

$$_1H^2 + {_2He^3} \quad = \quad {_2He^4} + {_1H^1} + 18.3MeV\ldots..(4)$$

The above equations can be combined as

$$6\,{_1H^2} + {_2He^3} + {_1H^3} = 2\,{_2He^4} + {_2He^3} + {_1H^3} + 2\,{_1H^1} + 2\,{_0n^1} + 43.2MeV \ldots(5)$$

5.13. Nuclear Fusion Reactor

Principle of Fusion reactor

Nuclear fusion is the principle of fusion reactor. To produce energy from hydrogen, the hydrogen must be raised to very high temperature (10 to 100 million degrees). Under this condition, the electrons will be stripped away from the nuclei, then there will be only electrically charged nuclei and free electrons. This is called **plasma**. Nuclei and electrons will be moving very rapidly in random directions.

If the plasma is hot enough, nuclei overcome the electrical barrier, undergo following reactions (1) and (2) and produce more energy, increasing the plasma's temperature further. This heat would then be used as in an ordinary power plant, to generate electricity.

$$_1H^2 + {_1H^2} = {_1H^3} + {_1H^1} \qquad \ldots(1)$$

$$_1H^2 + {_1H^2} = {_2He^4} + {_0n^1} \qquad \ldots(2)$$

The hot plasma cannot be kept in a container since it has high temperature. There are two approaches to confining the plasma

1. Magnetic confinement
2. Inertial confinement

1. **Magnetic confinement:** The trajectories of fast-moving electrically charged particles are bent in a magnetic field. Plasma physicists have devised very complex arrangements of magnets, so that the electrons and ions are kept within a finite volume, when a particle approaches the edge of the volume, it feels a magnetic force that turns it back into the volume.
2. **Inertial confinement:** In inertial confinement, a small pellet of solid hydrogen fuel is hit on all sides by many laser beams. This compresses the pellet and heats it to a fusion temperature. The pellet is quickly vaporized and begins to dissipate, but it may stay together long enough so that fusion can generate additional heat. The object in each of these techniques is not to keep the plasma confined forever, but to keep it confined long enough so that fusion can generate more heat energy than was used to trigger the fusion.

Construction and Working of fusion reactor

a) The fusion reactor heats fuel (Deuterium and Tritium) to form high-temperature plasma. It squeezes the plasma so that fusion can take place. The power needed to start the fusion reaction is about **70**

Megawatts, but the output power is **500 Megawatts**. The fusion reaction lasts from **300 to 500 seconds**. (Eventually, there will be a sustained fusion reaction).

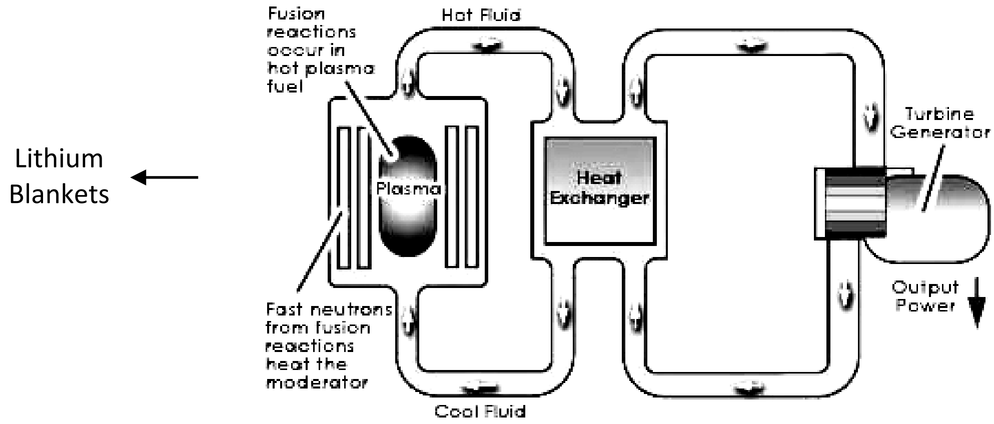

Fig.5.5. Fusion reactor

b) The lithium blankets outside the plasma reaction chamber absorb high-energy neutrons from the fusion reaction to make more Tritium fuel. The blankets also get heated by the neutrons as shown in Fig. 5.5.

c) The heat is transferred by a water-cooling loop to a heat exchanger to make steam.

d) The steam drives the electrical turbines to produce electricity.

e) The steam condenses back into water to absorb more heat from the reactor in the heat exchanger.

5.14. Nuclear power plant

Nuclear power plant is controlled self-sustaining chain reacting system supply nuclear energy. It is an experimental arrangement for converting nuclear energy into electrical energy.

Principle: Nuclear power plant is based on the principle of converting nuclear energy into electrical energy. Enormous heat is released during nuclear fission, the heat energy produced is used in producing steam which runs the steam turbine and electric generator is connected to the turbine to produce electrical energy.

Description and working: A schematic Nuclear power plant is as shown in Fig. 5.6 the parts of the nuclear power plant are

a) Nuclear reactor
b) Heat exchanger
c) Steam turbine
d) Electric generator
e) Condenser

a) Nuclear reactor: Nuclear fission reaction takes place in the reactor and enormous amount of energy is released.

b) Heat exchanger: The heat produced due to nuclear fission is given to the coolant. It carries away the heat to the heat exchanger and produce steam.

c) **Steam turbine:** The steam produced in the heat exchanger is passed on the steam turbine, which coverts it into mechanical energy.

d) **Electric generator:** It is connected to the steam turbine; here mechanical energy of the turbine is converted into electrical energy.

e) **Condenser:** It is used to condense the steam by reducing pressure of the expanded steam.

Advantages of Nuclear power plant

a) Space requirement of the nuclear power plant is less as compared to other conventional plant of equal size.

b) Fuel consumption is very less.

c) Fuel transportation cost is low and no storage of fuel is required.

d) Produce more power.

e) Plant is not affected by adverse weather condition.

Disadvantages of Nuclear power plant

a) Initial cost of nuclear power plant is high when compared with other conventional power plant of equal capacity.

b) Not suitable for varying conditions.

c) Radioactive waste should be disposed carefully; otherwise it may have bad effect on the health of workers and other persons.

d) Maintenance cost of the plant is high.

5.15. Classification of nuclear reactors:

Nuclear reactor may be classified under different ways they can be classified as

(a). Thermal reactors in which fission is induced primarily by thermal neutrons.

(b). Intermediate reactors in which fission is induced by epithermal neutrons.

(c). Fast neutron reactors in which fission is induced by fast neutrons.

Nuclear reactors can also be classified as

1. **Homogeneous reactors:**

   The nuclear fuel and the moderator represent a uniform mixture.

2. **Heterogeneous reactor:**

   Separate fuel rods are inserted in the moderator. Nuclear reactors may also be classified according to the type of the moderator, such as

   (i) Graphite reactor

   (ii). Heavy water reactor

3. **Breeder reactor:**

A reactor which is used to convert fertile material like $_{92}U^{235}$ into fissionable material is called the breeder reactor.

5.16. Nuclear reactor

A nuclear reactor is a device in which the nuclear fission reaction takes place in a self sustained and controlled manner. The first nuclear reactor was built in 1942 at Chicago USA.

Essential components of Reactor and its functions

Following are the essential components of a nuclear reactor.

a) Reactor core
b) Moderator
c) Reflector
d) Coolant
e) Control system
f) Shielding

a) Reactor core:

It is the heart of the reactor; here the nuclear fission takes place at a steady controlled rate. The fission energy is liberated in the form of heat.

The reactor core consists of fuel, moderator, control rods and the space for coolant. The reactor core is cylindrical in shape and is contained in a pressure vessel.

Reactor fuel: The commonly used fissionable materials are the Uranium isotopes, Thorium isotopes and the Plutonium isotopes. It is generally taken in the form of rods.

b) Moderator: It is a material, which reduces the speed of fast neutrons in the process of nuclear fission to thermal neutrons. The commonly used moderators are Heavy water, Ordinary water, Graphite and Beryllium oxide. The purpose of moderators is to reduce the energy of neutrons, but it should not absorb neutrons.

c) Reflector: It is a material, which is used to reflect back some of the escaping neutrons into the reactor core. It will increase the neutron flux in the interior of the core. The materials used for reflector is the same as the materials used for the moderators.

d) Coolant: The purpose of coolant is to remove heat produced inside the reactor core. The coolant is taken generally in the form of liquid or gas. It is circulated through the reactor core by means of a coolant pump. The following materials are used as coolants:

1. Water(light water or heavy water)
2. Gases (helium, carbon dioxide)
3. Liquid metals (sodium and organic liquid.

e) Control systems (Control Rods):

The control system is necessary in a nuclear reactor for the following purposes

1. To control the rate of fission.
2. To maintain the chain reaction at a steady state.
3. To shut down the reactor at the steady state.

The commonly used control rods are made up of elements like Boron or Cadmium. The control rods are inserted into the core and they pass through the space in between the fuel tubes and through the moderator. By pushing them in or pulling out, the reaction rate can be controlled.

f) Shielding

In a nuclear reactor, a large amount of penetrating radiations like α–rays, β-rays, in addition to neutrons are emitted. These radiations are very harmful to the technicians working around the reactor. Hence a thick shield, in the form of concrete walls of many meters thick, always surrounds a reactor

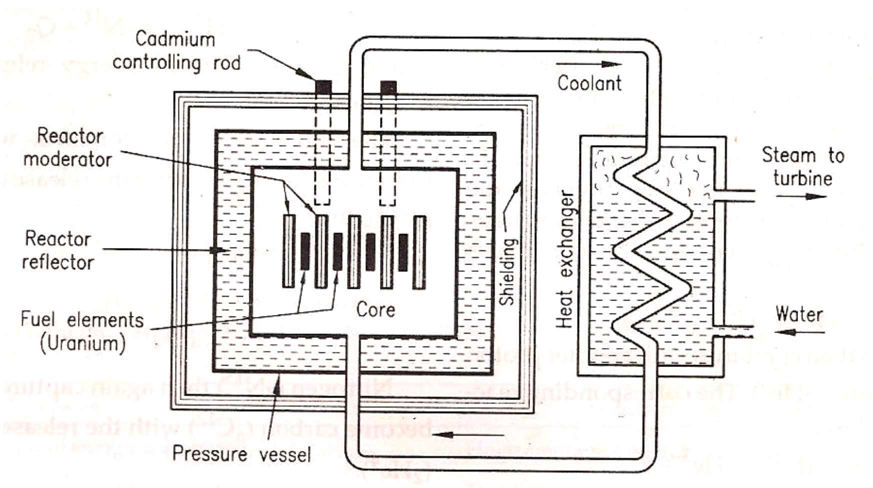

Fig.5.6. Nuclear reactor

Working of the reactor

Neutrons produced by action of alpha (α) particles on Polonium or Beryllium is slowed down and is used initially to cause fission of say U^{235} nuclei. Thus chain reaction sets in. The heat energy is produced during nuclear fission reaction. This heat energy is absorbed by the coolant and is used to drive the machinery producing electricity.

Power output of a nuclear reactor

The fission rate of a reactor i.e., total number of nuclei undergoing fission per second in a reactor is = $nv\sigma NV = \varphi\sigma NV$

Where, n= average neutron density i.e., number per m^3,

v= average speed in ms^{-1},

φ= nv = average neutron flux,

N= number of fissile nuclei per m^3,

σ= fission cross section in m^2,

V= volume of the nuclear fuel.

5.17. Thermal reactor:

In these reactors, the fast neutrons are slowed with the use of moderators. The slow neutrons are absorbed by the fissionable fuel and the chain reaction is maintained. The moderator is most essential component in these reactors. (The construction and working of these reactors are same as explained earlier).

Water cooled reactors:

In these reactors, the water is used both as a coolant and moderator. The water may be ordinary. i.e., “light water” or “Heavy water”.

Light water reactor (LWR):

In these reactors, the ordinary or light water is used both as a coolant and moderator, since it has a high hydrogen concentration.

There are two basic types of light water reactors

(i). Pressurized water reactor (PWR) and

(ii). Boiling water reactor (BWR)

5.18. PRESSURIZED WATER REACTOR (PWR)

Definition: It is a nuclear reactor in which water is used as moderator coolant and reflector and is kept under pressure to prevent it from boiling at the operating temperature.

Principle:

Huge amount of heat energy is released when a heavy nucleus undergoes a nuclear chain fission process. This heat energy is converted into steam which is used for the production of electricity.

Description:

The nuclear power plant using a pressurized water reactor is shown in Fig 5.7.

PWR uses two circuits.

1. A primary circuit containing the reactor, pressuriser , one side of
   a heat exchanger and coolant pump.

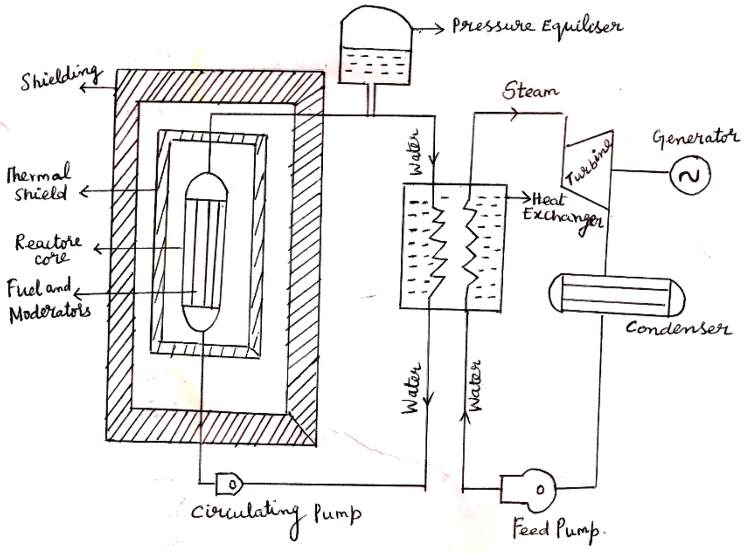

Fig 5.7.

2. A secondary circuit containing turbine, condenser, feed pump and the steam side of the heat exchanger.

Fuel

It uses enriched Uranium (contains more fissionable U^{235}) as fuel.

Moderator, reflector and coolant

Water (light or heavy) is used as moderator, reflector and coolant.

Pressure Equalizer

To maintain the constant pressure in the primary circuit throughout the load range, the pressurizing tank or pressurizer is included in the circuit. It is simply a pressure vessel with a heater at the bottom and a water spray at the top of the vessel and is filled with steam which maintains the primary pressure circuit.

If the primary circuit pressure drops the heater is operated which generates steam and increases the pressure of the primary circuit. If the primary circuit pressure becomes too high, cold water is sprayed into the steam in the pressurizer. This condenses the steam and reduces the primary circuit pressure.

Heat exchanger or steam generator

In the heat exchanger heat is carried by the coolant and is converted into steam.

Turbine

It converts heat energy into mechanical energy.

Electric generator

Here, mechanical energy of the turbine is converted into electrical energy.

Condenser

It is used to condense the steam by reducing the pressure of the expanded steam.

Circulating pump

It is used to circulate the water (coolant)

Feed pump

It is used to pump the condensed steam from the condenser to reactor core.

Operation

The water (coolant) in the primary circuit is pumped to the reactor core. The coolant is heated by the heat energy released due to nuclear fission in the reactor core. The hot water passes through the heat exchanger. In the heat exchanger, the coolant transfer the heat energy to feed water and steam is generated. The steam turbine is connected to the generator for the production of electrical power. The water from the heat exchanger is again circulated by the coolant pump.

To prevent boiling of water in the reactor it is kept under a pressure of 14 MN/m^2. This enables water to carry more heat from the reactor. The pressuriser regulates the pressure in the circuit to safe guard the reactor from the power fluctuation.

Advantage:

1. Water which is cheaply available in plenty is used for coolant, moderator and reflector.
2. It is compact in size and has high power density (65 KW/litre)
3. Number of control rods required is less.
4. Steam is not contaminated by radiation. Hence maintenance of turbine is normal.
5. It gives complete freedom to inspect and maintain the turbine, feed heaters and condenser during operation.

Disadvantage:

(i) Since there is high primary circuit pressure, it requires strong pressure vessel. Hence it needs high capital cost.

(ii) Thermal efficiency of the plant is low since low pressure is maintained in the secondary circuit

(iii) Corrosion problem occurs at high pressure and high temperature of water. Hence proper vessel is necessary to prevent corrosion. It further increases the cost of the plant.

(iv) Reprocessing of fuel is difficult since it is affected by radiation.

(v) For changing the fuel, it is necessary to shutdown the reactor. It requires more than one or two months.

5. 19. BOILING WATER REACTOR (BWR)

Definition:

It is a nuclear reactor in which water in contact with fuel elements is used as coolant, moderator and reflector.

Principle:

Huge amount of heat energy is released when a heavy nucleus undergoes a nuclear fission chain reaction. This heat is converted into stream which is used in the production of electricity.

Description:

The nuclear plant using boiling water reactor is shown in Fig 5.8. It consists of reactor, turbine, condenser and feed pump.

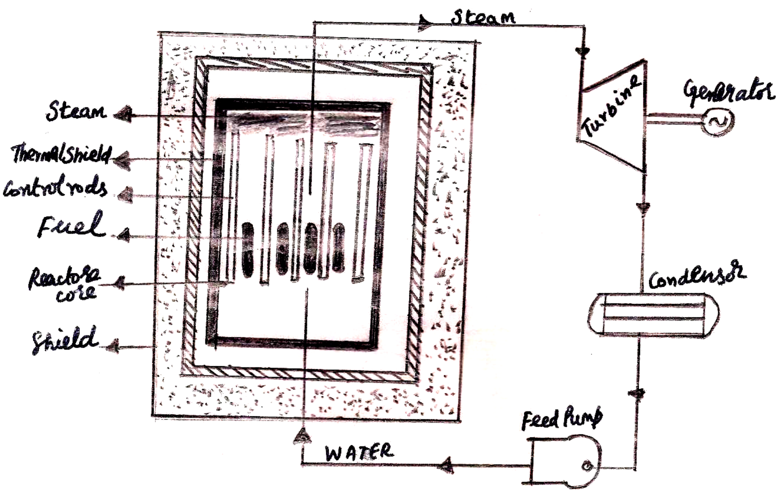

Fig 5.8. Boiling Water Reactor

Reactor:

Nuclear fission chain reaction takes place and huge amount of heat energy is released in the reactor.

Fuel:

It uses enriched Uranium as fuel.

Coolant, Moderator & reflector:

Water is used as coolant, moderator and reflector as in the PWR.

Turbine:

It converts heat energy into mechanical energy.

Electric generator:

Mechanical energy is converted into electrical energy.

Condenser:

It is used to condense the steam by reducing the pressure of the expanded steam.

Feed Pump:

It is used to pump the condensed steam from the condenser to the reactor core.

Working:

Water (coolant) is pumped into the reactor at the bottom. This water is heated by the heat released due to the nuclear fission in the reactor core and gets converted into steam. The steam leaves from of the top the reactor to the turbine is connected to the generator for the production of electrical power. Exhaust steam from the turbine passes through the condenser and condensed. The condensed steam is re-circulated again using feed pump.

Advantages:

1. As water is allowed to boil inside the reactor, the pressure inside the reactor is small. Hence high pressure vessels are not required. It reduces the cost considerably.
2. Steam generator, pressuriser, circulating pump and connecting piping are not required. Hence the cost is further reduced.
3. Thermal efficiency of BWR plant is more than PWR.
4. It is a self controlled reactor as the reactivity automatically is reduced if the vapor is not dense enough to moderate the neutrons effectively.

Disadvantages:

1. It cannot meet a sudden increase in power demand.
2. The steam leaving the turbine is slightly radioactive. Hence shieldingof turbine and piping is essential.
3. The size of the vessel is large since it has low power density. There is a possibility of burn out of fuel in the reactor as boiling of water on the surface of fuel is allowed

5.20. FAST BREEDER TEST REACTOR:

The below figure 5.9 shows a fast breeder reactor system. In this reactor the core containing U^{235} surrounded by a blanket (a layer of fertile material outside the core) like U^{238}. In this reactor no moderator is used. The fast moving neutrons liberated due to fission U^{235} are absorbed by U^{238} which gets converted into fissionable material Pu^{239} which is capable of sustaining chain reaction.

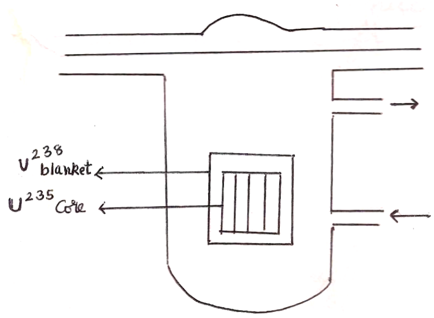

Fig 5.9

Thus this reactor is important because it breads fissionable material from fertile material U^{238} available in large quantities. Fast breeder reactors are better than conventional reactor from the point of view of safety and thermal efficiency.

For India which already is fast advancing towards self reliance in the field of nuclear power technology the fast breeder reactor becomes important in view of massive reserves of Thorium and the finite limits of its Uranium resources. The research and development efforts in the fast breeder reactor technology will have to be stepped up considerably if nuclear power generation is to make any impact on the countries total energy needs in the distant future.

5. 20.1 LIQUID METAL FAST BREEDER REACTOR

Characteristics:

Fuel	:	Mixed Fuel (PUO and UO)
Coolant	:	Liquid Sodium
Moderator	:	No moderator is used
Condenser	:	It is used to condense the steam by reducing the pressure of the steam.
Generator	:	Mechanical energy of the turbine is converted into Electrical Energy.
Power output:		40 MW thermal and 13.2MW electrical.

Fast breeder reactors are designed to create or breed new fissile material for producing useful electrical power. Most produce fissile Plutonium from U^{238}. The fuel rods in the core region contain a mixture of fissile Pu^{239} and U^{238}. The active core region is surrounded by a blanket of fertile U^{238}. This blanket region captures neutrons that would otherwise be lost through leakage, it produces additional fissile material. A fast neutron reaction with U^{238} producing Pu^{239} is shown below.

$$_{92}U^{238} + {}_{0}n^{1} \longrightarrow {}_{92}U^{239} \xrightarrow{2\beta} {}_{94}Pu^{239}$$

Similarly from $_{90}Th^{232}$, we can produce fissionable $_{92}U^{233}$

$$_{90}Th^{232} + {}_{0}n^{1} \longrightarrow {}_{90}Th^{233} \xrightarrow{2\beta} {}_{92}U^{233}$$

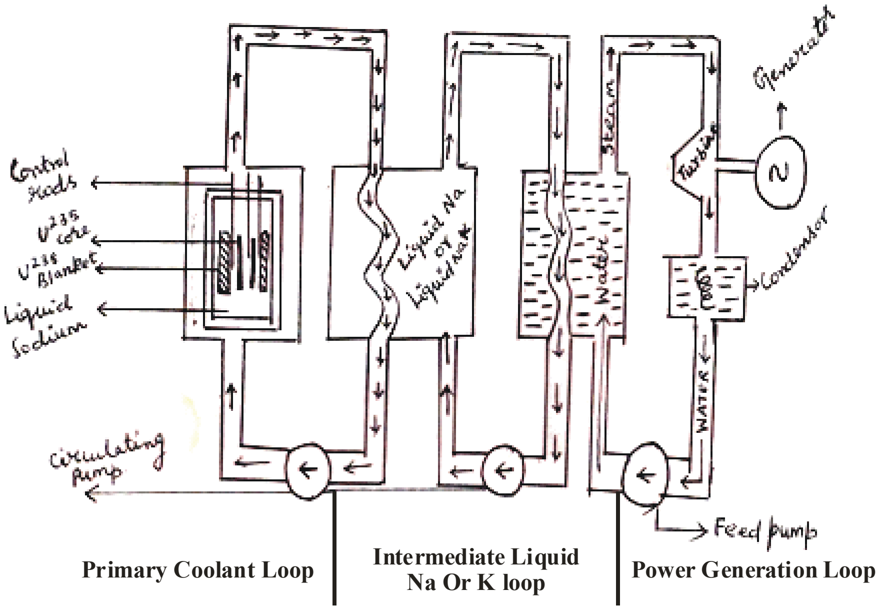

Fig. 5.10 FBTR Reactor

By suitable controlled conditions more $_{94}Pu^{239}$ can be produced in the natural Uranium blanket then U^{235} that is used up in the reaction. For this reason this type of reactor is called fast breeder reactor. Figure 5.10 shows the parts of FBTR reactor.

In October 1985, the Indira Gandhi centre for Atomic Research (IGCAR) and Bhabha Atomic Research centre (BARC) jointly designed, constructed and operated the reactor. The FBTR is liquid metal fast breeder reactor. The reactor uses Plutonium- Uranium mixed Carbide fuel and liquid sodium as coolant.

The fuel is an indigenous mixture of 70% Plutonium Carbide and 30% Uranium Carbide. The reactor was designed to produce 40MW thermal power and 13 MW electrical powers. During the operation of the reactor heat is produced and this heat is minimized using proper coolant. A coolant with excellent heat transfer properties is required to minimize the temperature drop from the fuel surface to the coolant and also it must be non-moderating, considering the above reasons water is ruled out.

The coolants for fast breeder reactor are

1. Liquid Metal (Na or NaK)
2. Helium (He)
3. Carbon-di-oxide

The best coolant for fast breeder reactor is sodium in liquid because of the following reasons.

1. It has low absorption cross-sectional area.
2. It possess good heat transfer properties and low pressure property.
3. It does not react to any of the structural material in primary circuits.

4. The boiling point of sodium at atmospheric pressure is very high, so it can use unpressurized in a reactor.

The heat produced in the core is removed by two hydraulically coupled primary sodium loop which in turn transfer the heat to two independent secondary loops containing Sodium Potassium alloy. The secondary coolant while flowing through the intermediate heat exchanger transfers its heat to feed water. Proper heat insulation is provided on all pipelines to maintain sodium in liquid form.

Liquid sodium is more effective as a medium for heat transfer, but it a more reactive and undergoes radioactive decay by capturing a neutron.

$$_{11}Na^{23} + {}_{0}n^{1} \longrightarrow {}_{11}Na^{24}$$

$$_{11}Na^{24} \xrightarrow{\text{148 hours } (\beta)} {}_{12}Mg^{24} + {}_{-1}e^{0}$$

Because of the induced radioactivity of liquid Sodium an intermediate loop also using Na or NaK as coolant is between the primary loops and will act as a radioactive coolant.

Advantages:

1. The high concentration of fissile fuel and the absence of moderator make core of a fast reactor smaller than the thermal reactor of same power.

Merits of the Fast breeder reactors:

1. Extraction of 100 times more energy from Uranium than via light water reactors, enabling the use of depleted Uranium which is regarded as an unstable waste at present.
2. Much higher power generation than PHWR:

a. PHWR : 73 tons → 15 MW

b. FBR : 73 tons → 350 MW

3. Lower pollution factor
4. Much higher Thermodynamic efficiency (40% vs. 30%)

Pondicherry University 2 Marks Question and Answers

Part - A

1. What are prompt neutrons? (May 2012)

In nuclear fission reactions, neutrons emitted instantaneously by a nucleus undergoing fission are called prompt neutrons. These prompt neutrons are released directly from fission within 10^{-13} seconds of the fission event

2. **What is mass defect? (May 2013, 2017, Jan 2014, 2016)**

The difference between the actual mass of the nucleus and the sum of the masses of constituent nucleons is called mass defect.

3. **What is the principle used in D-D nuclear reactor? (Jan 2014)**

Thermonuclear reactions fusion reaction is the principle used in D-D nuclear reactor.

4. Determine the energy equivalent of 1 a.m.u. (Jan 2011), (May 2011), (May 2012)

The energy equivalent of 1 amu is 931 MeV.

5. What do you mean by nuclear fusion reaction? (Jan 2011, May 2017)

Nuclear fusion is a nuclear reaction in which two lighter nuclei fuse together to form a heavier and stable nucleus with the release of large amount of energy.

$$_1H^2 + {_1H^2} \longrightarrow {_2He^4}$$

For example when the two neutrons are brought together to form single Helium (α-particle) the following reaction takes place with the liberation of 24 MeV energy.

6. What is binding energy? (Jan 2013, 2016)

The amount of energy required to break up the atom into its constituents is equal to the mass defect the energy equivalent of mass defect is a measure of binding energy of the atom.

$$B.E = \Delta mc^2 = [z\, m_p + Nm_n - m]\, c^2$$

The binding energy is expressed in MeV using the relation.

$$1amu = 931.478\ MeV$$

7. The process in which nucleus of a heavy atom breaks into two, more or less equal segments with release of energy is known as-(d) Fission (Jan 2013)

a) Fusion b) Ionization c) radioactivity d) fission

8. Define the term multiplication factor. (May 2011)

Multiplication factor is the ratio of number of neutrons produced in any particular generation to number of neutrons produced in the preceding generation

$$K = \frac{\text{number of neutrons produced in any particular generation}}{\text{number of neutrons produced in the preceding generation}}$$

When k=1, fission chain reaction will be critical or steady

When k>1 it will be "building up" or super critical.

When k<1 it will be dying down or sub critical.

9. What are the essential components of a nuclear reactor?(May 2013)

1. Reactor core
2. Reflector
3. Moderator
4. Coolant
5. Control material
6. Shielding.

10. Explain the general properties of nucleus (Jan 2016)

(i). Nuclear size

If the nucleus is assumed to be spherical, an empirical relation is found to hold good between the radius of the nucleus R and its mass number A. It is given by

$$R \propto A^{1/3}$$

$$R = r_o A^{1/3} \qquad \ldots(1)$$

where r_o is the constant of proportionality and is equal to 1.3F (1 Fermi, F = 10^{-15} m)

(ii). Nuclear density

The nuclear density ρ_N can be calculated from the mass and size of the nucleus.

$$\rho_N = \frac{\text{Nuclear mas}}{\text{Nuclear volume}}$$

Nuclear mass = Am_N where, A = mass number and m_N = mass of one nucleon and is approximately equal to 1.67×10^{-27} kg.

$$\text{Nuclear volume} = \frac{4}{3}\pi R^3$$

$$= \frac{4}{3}\pi (r_o A^{1/3})$$

$$= \frac{Am_N}{\frac{4}{3}\pi\ (r_0 A^{1/3})^3} \qquad \ldots(1)$$

$$\therefore \quad \rho_N = \frac{m_N}{\frac{4}{3}\pi\, r_0{}^3}$$

Substituting the known values, the nuclear density is calculated as 1.816×10^{17} kg m^{-3} which is almost a constant for all the nuclei irrespective of its size.

(iii). Nuclear charge

The charge of a nucleus is due to the protons present in it. Each proton has a positive charge equal to 1.6×10^{-19} C.

∴ The nuclear charge = Ze, where Z is the atomic number.

(iv). Nuclear mass

As the nucleus contains protons and neutrons, the mass of the nucleus is assumed to be the mass of its constituents. Assumed nuclear mass = $Zm_P + Nm_n$, where m_p and m_n are the mass of a proton and a neutron respectively. However, from the measurement of mass by mass spectrometers, it is found that the mass of a stable nucleus (m) is less than the total mass of the nucleons. i.e., mass of a nucleus, $m<(Zm_p+Nm_n)$, $Zm_p + Nm_n - m = \Delta m$, where Δm is the mass defect .

(v) Spin angular momentum

Both the proton and neutron, like the electron, have an intrinsic spin. The spin angular momentum is computed by $L_s = \frac{l(l+1)h}{2\pi.}$

(vi). Resultant angular momentum

In addition to the spin angular momentum, the protons and neutrons in the nucleus have an orbital angular momentum. The resultant angular momentum of the nucleus is obtained by adding the spin and orbital angular momentum of all the nucleons within the nucleus. This total angular momentum is called nuclear spin.

11. What are the required specifications for the reflector? Mention the example.(Jan 2016)

Reflector should have

1. Materials of high scattering cross- section.
2. Low absorption cross section.

Materials used:

1. Heavy water 2. Water 3. Graphite 4.Berylliunm oxide.

12. Difference between nuclear fission and nuclear fusion.(Jan16,May17)

S.No.	Nuclear fission	Nuclear fusion
1.	Huge amount of energy is liberated	Huge amount of energy is liberated
2.	A heavy nucleus is split up in to two nuclei.	Two lighter nuclei are fused together.
3.	This process is possible even at room temperature	This process is possible only at very high temperature
4.	The link of this process are neutrons	The link of this process are protons
5.	The energy per nucleon is 0.85 MeV	The energy per nucleon is 6.75MeV

Additional 2 Marks Questions & Answer

1. Define nuclear fission with example (Jan 2016).

Nuclear fission is a nuclear reaction in which heavy nuclei after capturing a neutron splits up into two higher nuclei or nearly equal masses.

Here heavy nucleus uranium (z-92) splits up into two higher nucleus (Barium (z=56), Krypton (z=36)) after capturing a neutron.

2. What is nuclear energy?

When a heavy nucleus undergoes a fission process a large amount of energy is released together with the emission of neutrons which further produce fission this energy is called nuclear energy or atomic energy.

3. What is chain reaction? (Jan 2016)

Chain reaction is a process in which the nuclear fission of an atom induces nuclear fission in another atom which again induces fission in another and so on.

It is self-programming process in which number of neutrons goes on multiplying rapidly almost in geometrical progression during fission till whole of fissionable material is disintegrated.

4. For nuclear fission neutrons are used similarly for nuclear fusion which particle is used?

For nuclear fusion proton is used.

5. What is thermo-nuclear reaction?

Thermo-nuclear reaction is a type of nuclear fusion reaction takes place at very high temp (10^9 °C) in which two lighter nuclei fuse together to form a heavier and stable nucleus with the release of large amount of energy.

6. What is stellar energy?

The temperature of the sum is very high ($3X10^7$ °C) it radiates tremendous amount of energy ($3.8\ X10^{26}$ joules of energy per second).The energy produced in the sun and star is known as stellar energy.

7. What is proton –proton cycle?

Proton –proton cycle is a chain of nuclear fusion reactions assumed to be responsible for production of energy in the sun the reactions lead to the conversion of hydrogen into helium and are given by

$${}_1H^1 + {}_1H^1 \longrightarrow {}_1H^2 + {}_1e^0 + Q_1$$

$${}_1H^2 + {}_1H^1 \longrightarrow {}_2He^3 + {}_1e^0 + \gamma$$

$${}_2He^3 + {}_2He^3 \longrightarrow {}_2He^4 + 2\ {}_1H^1 + Q_2$$

$$4{}_1H^1 \longrightarrow {}_2He^4 + 2 + {}_1e^0 + Q$$

8. What is carbon –nitrogen cycle? (May 2017)

Carbon nitrogen cycle is a cycle of nuclear reactions resulting into the conversion of hydrogen nuclei into helium nucleus carbon acting as a catalyst with the liberation of a large amount of energy per cycle .energy liberation in sun and other stars is supported to take place via carbon –nitrogen cycle.

$$_6C^{12} + {}_1H^1 \longrightarrow {}_7N^{13} + Q^1$$
$$_7N^{13} \longrightarrow {}_6C^{13} + {}_{+1}e^0$$
$$_6C^{13} + {}_1H^1 \longrightarrow {}_7N^{14} + Q^2$$
$$_7N^{14} + {}_1H^1 \longrightarrow {}_8O^{15} + Q^3$$
$$_8O^{15} \longrightarrow {}_7N^{15} + {}_{+1}e^0$$
$$_7N^{15} + {}_1H^1 \longrightarrow {}_6C^{12} + {}_2He^4$$
$$4{}_1H^1 \longrightarrow {}_2He^4 + 2\ {}_{+1}e^0 + 27\ MeV$$

9. What is nuclear reactor?

Nuclear reactor is a device or apparatus in which nuclear fission is produced under a self-sustaining controlled nuclear chain reactions it is a controlled self- sub staining chain reacting system supplying nuclear energy.

10. How are the nuclear reactors classified according to the type of moderator?

1. Graphite reactor.
2. Heavy water reactor.

11. How are the nuclear reactors classified on the basis of neutrons?

1. Thermal reactors in which nuclear fission is induced primarily by thermal neutrons.
2. Intermediate reactors in which nuclear fission is induced by epithermal neutrons.
3. Fast reactors in which nuclear fission is induced by fast neutrons.

12. What is breed reactor?

Breed reactor is a nuclear reactor which is used to convert fertile material like $_{92}U^{235}$ in to fissionable material.

13. What is a reactor core in a nuclear reactor?

Reactor core is the main part of reactor which contains the fissionable material called the reactor fuel. The region in which the fuel is kept is called reactor core, here the nuclear reaction takes places and a huge amount of heat is generated.

14. What are the common fissionable materials used in the nuclear reactor?

The commonly used fissionable materials are

Uranium isotope U^{233}, U^{235}, U^{236}

Thorium isotope Th^{232}

Plutonium isotope Pu^{239}, Pu^{240}, Pu^{241}

15. What is the functioning of a reflector in the nuclear reactor?

Reflector in a nuclear reactor is usually placed around the reactor core. Its function to reflect back some of the neutrons that leak out from the surface of the core and to increase the neutron flux in the interior of the core.

16. What is moderator?

Moderator is a substance which is used in a nuclear reactor to slow down the fast neutrons in the process of fission to thermal neutrons.

Commonly used moderator

1. High boiling point
2. Large scattering cross section
3. Small absorption cross section
4. Low atomic number.

Materials used:

1. Heavy water 2. Water
3. Graphite 4.berylliunm oxide.

17. What is the role of a coolant in a nuclear reactor?

Coolant removes the intense heat produced in the reactor core during nuclear fission. The coolant is generally pumped through the reactor in the form of liquid or gas it is circulated throughout the reactor so as to maintain a uniform temperature.

18. What is the required specification for coolant?

Coolant should have the following properties

1. High boiling point.
2. High specific heat.
3. Easy in pumping
4. Chemically stable
5. Cheap

19. Mention the functioning of a control material and example.

Nuclear fission process can be controlled by controlling the number of neutrons in it. Control material simply absorbs excess of neutrons, it is achieved by introducing neutron absorbing material called control rods and by adjusting their position in the core.

20. What is the required specification for the control material and give example.

Control rods should have a large neutron absorption cross section

Materials used 1. Boron rod 2. Cadmium rod.

21. Why is shielding necessary in a nuclear reactor?

In an nuclear reactor various types of rays are emitted these rays are dangerous to the persons working near the reactor hence thick walls of cement and concrete are constructed around the reactor they are known as shields.

22. What is water cooled reactors?

Water cooled reactors are nuclear reactors in which cooling is done by water. The water is used both as coolant and moderator. The water may be ordinary i.e., light water or heavy water.

23. What is light water reactor (LWR)? What are its types?

In these reactors the ordinary or light water is used both as a coolant and moderator.

There are two basic types of light –water reactors.

1. Pressurized water reactor (PWR)
2. Boiling water reactor (BWR)

24. What is pressurized water reactor (PWR)?

Pressurized water reactor (PWR) is a nuclear reactor in which water used as a moderator, coolant and reflector coolant and reflector is kept under pressure to prevent it from boiling at the operating temperature.

25. What is Boiling water reactor (BWR)?

Boiling water reactor (BWR) is a nuclear reactor in which water is used as both coolant and moderator hence water is allowed to boil inside the reactor core itself and it is converted into steam.

26. What is gas cooled reactor?

A nuclear reactor in which cooling of the core is achieved by circulation of a gas carbon dioxide (CO_2) Gas is used as coolant graphite as moderator and natural Uranium as fuel.

27. What are conditions to be satisfied for sustained chain reaction?

For sustained chain reaction the following conditions are

1. the chain reaction can be sustained only when each fissioned nucleus produces on the average at least one secondary neutron which cause further fission this is possible only if multiplication factor (k)=1, but if k<1, the neutron population decreases and chain reaction fails to develop.
2. The proportion of U^{235} to U^{238} should be increased so as to increase the probability of fission capture in neutrons by U^{235}.
3. There should be a moderator to slow down the neutrons to thermal energy without absorbing them.
4. The materials used a moderators and the uranium sample should be over 99.99% pure in order to avoid non-fission capture of neutrons by impurities.
5. There is a certain size called the critical size for the system.

28. What is critical size and critical mass of fissionable material?

Critical size is defined as the minimum size for which the number of neutrons produced in the fission process just balance those lost by leakage and non-fission capture the mass of the fissionable material at this is called the critical mass if the size is less than the critical size a chain reaction is not possible.

29. What is the efficiency of a nuclear reactor?

Efficiency of a nuclear reactor is defined as the ratio of heat given to the heat exchanger to heat gained due to nuclear fission.

$$\text{Effiency } (\eta) = \frac{\text{Heat given to the heat exchanger}}{\text{Heat gained due to nuclear fission}}$$

30. What is meant by thermal neutron? Point out its importance.

Thermal neutrons are the slow moving neutrons of energy 0.0253 eV corresponding to the velocities of 2200m/s. These thermal neutrons are responsible for producing nuclear fission.

PROBLEMS

1. **A neutron breaks into a proton and an electron. Calculate the energy produced in this reaction in MeV. Mass of an electron = 9.1 X 10^{-31} kg, mass of proton = 1.6747 X 10^{-27}kg and speed of light = 3 X 10^8m/sec**

   Solution:

   Mass defect (Δm)= mass of neutron - (mass of proton + mass of neutron)

   $(\Delta m) = 1.6747 \text{ X } 10^{-27} – (1.6747 \text{ X } 10^{-27} + 9 \text{ X } 10^{-31})$

   $= 0.0013 \text{ X } 10^{-27}\text{kg}$

   $E = \Delta mc^2 = 0.0013 \text{ X } 10^{-27} \text{ X } (3 \text{ X } 10^8)^2 \text{ Joule}$

   $= 0.0013 \text{ X } 10^{-27} \text{ X } (3 \text{ X } 10^8)^2 / 1.6 \text{ X } 10^{-13} \text{ MeV}$

   $E = 0.73\text{MeV}$

2. **Find the amount of energy produced in Joules due to fission of 1gram of Uranium 235 in one second assuming that energy released per fission is 200MeV.**

   Solution:

   Amount of energy produced due to fission (or) Power output from the source $P = \frac{mNE}{AT}$

   m - mass of the fuel
   N- Avogard's number
   E- Energy released per fission in Joules
   A- Atomic mass of the fuel
   T- Time in sec

$$P = \frac{1 \times 10^{-3} \times 6.02 \times 10^{26} \times 200 \times 10^{6} \times 1.6 \times 10^{-19}}{235 \times 1}$$

   P = 8.1974 X 10^{10} Joules

3. **Calculate the energy released by the fission of 2gm of $_{92}U^{235}$ in KWh. Given that the energy released per fission is 200MeV.**

Solution:

Amount of energy produced due to fission (or) Power output from the source $P = \frac{mNE}{AT}$

m - mass of the fuel

N- Avogard's number

E- Energy released per fission in Joules

B- Atomic mass of the fuel

T- Time in sec

$$P = \frac{2 \times 10^{-3} \times 6.02 \times 10^{26} \times 200 \times 10^{6} \times 1.6 \times 10^{-19}}{235 \times 60 \times 60 \times 60 \times 10^{3}}$$

P = 4.55 X 10^4 KWh

4. **Calculate the energy released by the fission of 10Kg of $_{92}U^{235}$ per day. Given that the average energy released per $_{92}U^{235}$ fission is 200MeV.**

Solution:

Amount of energy produced due to fission (or) Power output from the source $P = \frac{mNE}{AT}$

m - mass of the fuel

N- Avogard's number

E- Energy released per fission in Joules

C- Atomic mass of the fuel

T- Time in sec

$$P = \frac{10 \times 6.02 \times 10^{26} \times 200 \times 10^{6} \times 1.6 \times 10^{-19}}{235 \times 24 \times 60 \times 60}$$

P = 9.48 X 10^9 Watt

Pondicherry University (Part – B Questions)

1. **Explain the construction and working of a Pressurized Heavy Water Reactor and boiling water reactor. (Jan 2014), (Jan 2013, 2016), (May2011, 2012,2013, 2017)**

   Answer: Refer Page No:16 to 18

2. **Discuss in detail about the nuclear fusion reactors. (Jan 2014)**

   Answer: Refer Page No:11 to 13

3. **Explain the general properties of Nucleus on the basis of size, mass and density. (Jan 2014), (Jan 2013), (May 2013)**

   Answer: Refer Page No:2 to 4

4. **(a). Explain the classification of nuclear reactors. (May11,12,17)**

   Answer: Refer Page No:13 to 14

   (b) Explain the construction and working of a thermal reactor. (May2011), (May 2012)

   Answer: Refer Page No:14 to 16

5. **Discuss in detail the various materials used in nuclear reactors.**
   (Jan 2016)

   Answer: Refer Page No:14 to 16

6. **Explain how D-D and D-T reactions take place in fusion.(Jan 2016)**

   Answer: Refer Page No:10

7. **Explain the construction and working of FBTR.**

   Answer: Refer Page No:20 to 23

8. **Explain the nuclear fusion reaction which takes place in the sun and stars.**

   Answer: Refer Page No:9 to 10

9. **Explain how chain reaction takes place in Fission.**

   Answer: Refer Page No:7 to 8

10. **(a) How many Kilowatt hour of energy will be released by the complete fission of 1gm of U^{235}? Give that the energy released per fission is 200 MeV.**

    Answer: Refer Page No:32

(b) Find the binding energy of an alpha particle from the following data. Mass of the He nucleus = 4.001265 a.m.u, Mass of the Proton = 1.007277 a.m.u, Mass of the Neutron = 1.008666 a.m.u.

Solution:

Alpha particle of Helium consists of two protons , two neutrons and two electrons ($_2He^4$).

Mass of a proton = 1.007277 a.m.u.

Mass of two protons = 2 x 1.007277= 2.01456 a.m.u

Mass of a neutron = 1.008666 a.m.u

Mass of two neutrons = 2 x 1.008666 = 2.01732 a.m.u

So, Mass of the parts of the alpha particle ($_2He^4$) = 4.03188 a. m.u

We know mass of an alpha particle = 4.00153 a.m.u

Δm = 0.03035 a.m.u

Now, 1 a.m.u = 1.66054 x 10^{-27} kg

So, Δm = 0.03035 x 1.66054 x 10^{-27} kg = 0.05040 x 10^{-27} kg

So, binding energy is BE = $\Delta m\, c^2$ = 0.05040 x 10^{-27} x $(3 \times 10^8)^2$

= 4.536 x 10^{-12}J

Let us express this in terms of eV,

1eV = 1.6 x10^{-19} J

So, BE = $\frac{4.536\, X\, 10^{-12}}{1.6\, X\, 10^{-19}}$ = 2.835 X 10^7eV

BE of alpha particle = 28.35 MeV

11. Discuss with neat sketch the construction and working of Boiling Water Reactor (BWR) (May 2017).

Answer: Refer Page No:19 to 20